What Breaks Monocular SLAM in Microgravity?

An Initial Benchmark on Rotation-Dominant Astrobee ISS Sequences

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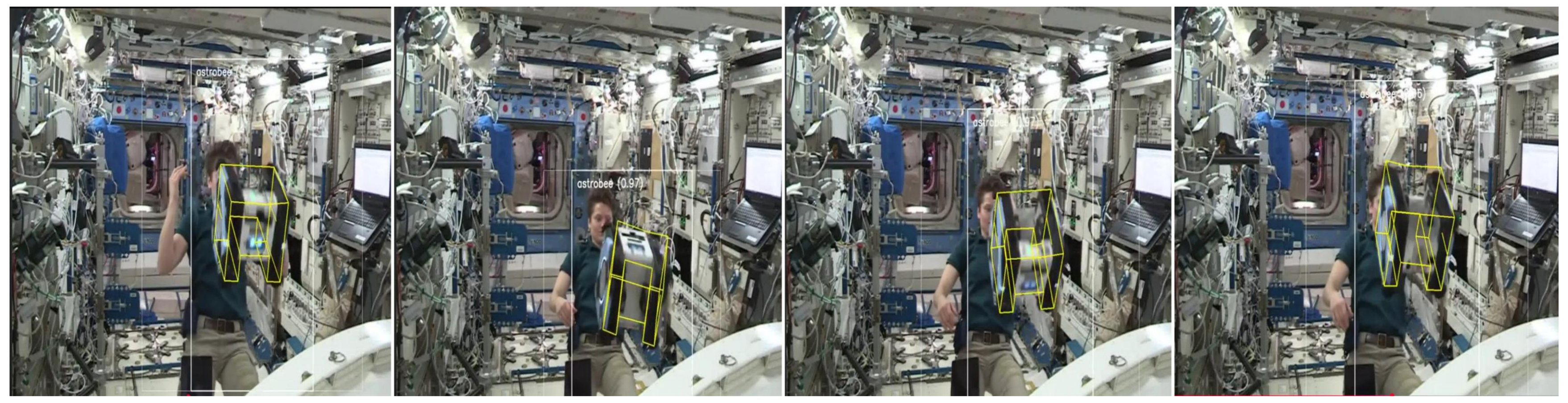
Motivation



Free-flyers undergo **unrestricted 360° rotation**

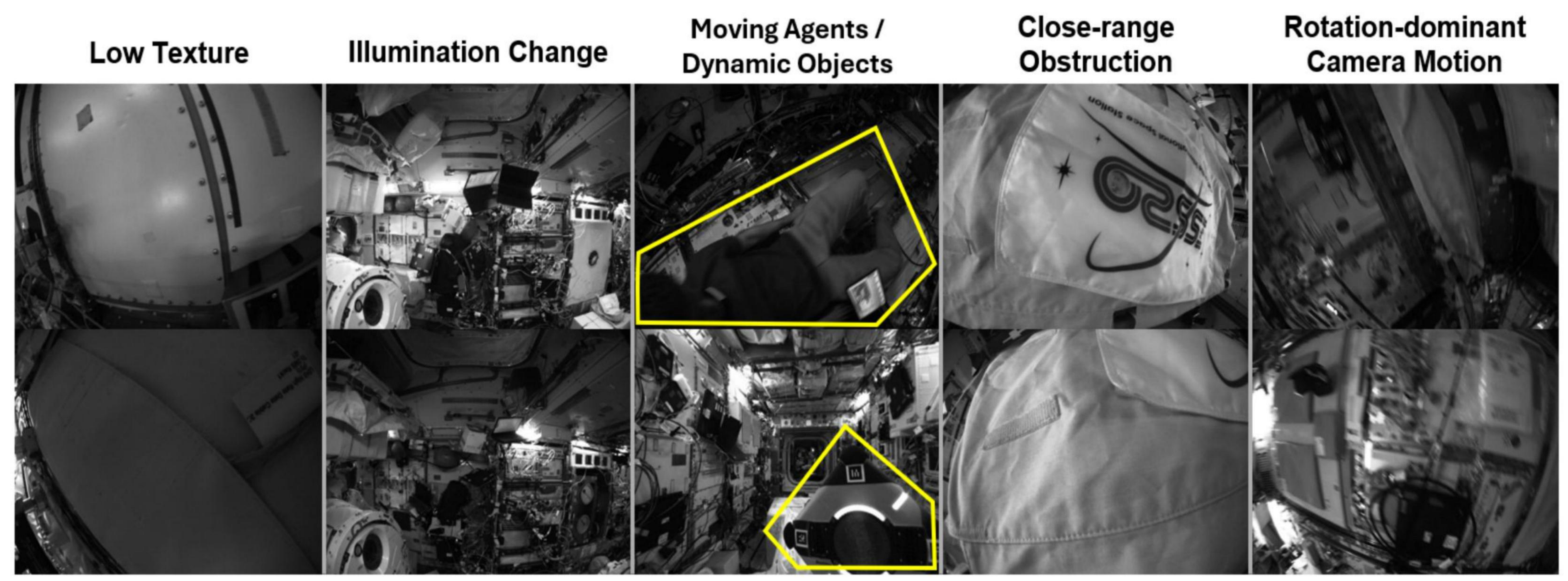
- **Failure mechanism:** 360° rotation removes mapped landmarks, and blur breaks local correspondences.
- **Operational impact:** this degrades AstroLoc2 onboard localization.

Contributions

1 **Visual challenge taxonomy** for Astrobee ISS imagery as a challenge-aware benchmark framework.

2 **Challenge-oriented benchmark** comparing 3DGS SLAM and geometric foundation models on rotation-dominant ISS sequences.

Visual Challenge Taxonomy for Astrobee Datasets



Five visual challenge categories — this study focuses on the **rotation-dominant subset** (rightmost).

Methods & Setup

Dataset: Astrobee ISS Free-Flyer Dataset
Seq.: iva_kibo_rot, ff_return_rot

Mechanism view: ***prior* × *coupling***
for rotation-dominant robustness

Baselines

3DGS-Based SLAM

MonoGS
No priors / no coupling.

HI-SLAM2
Depth-normal priors + pose-graph BA.

WildGS-SLAM
DINOv2 features + uncertainty-weighted BA.

Foundation-Model-Based SLAM

MUSt3R
Feed-forward pointmaps only.

MASt3R-SLAM
Pointmaps + tracking + BA.

Quantitative Results on Astrobee ISS Dataset

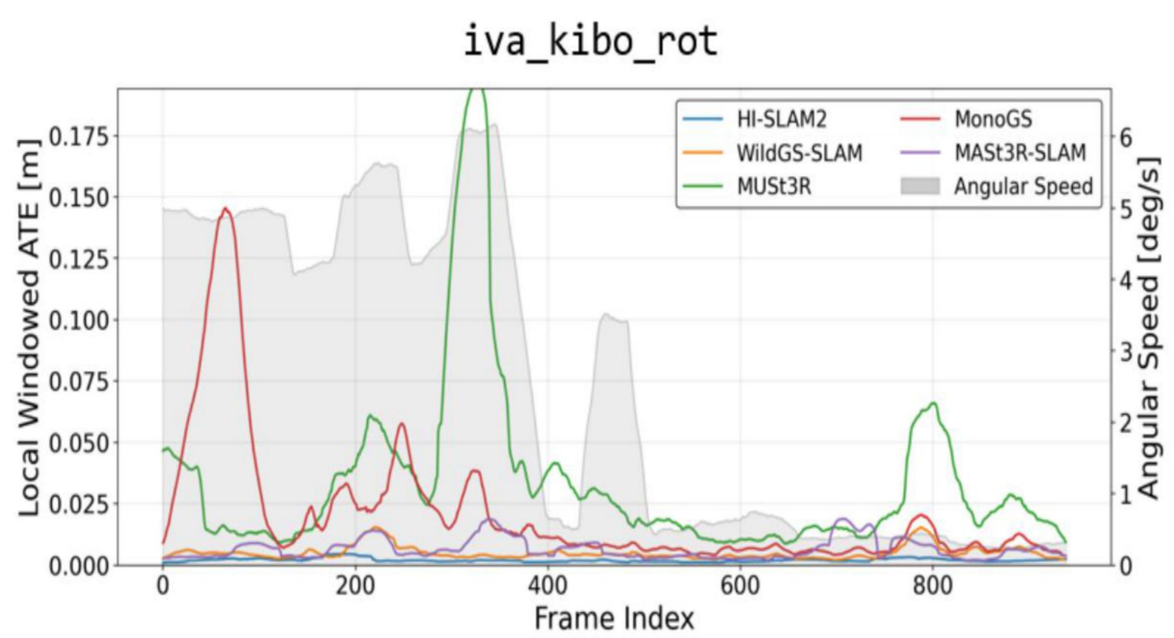
Table I. Global ATE RMSE [m]

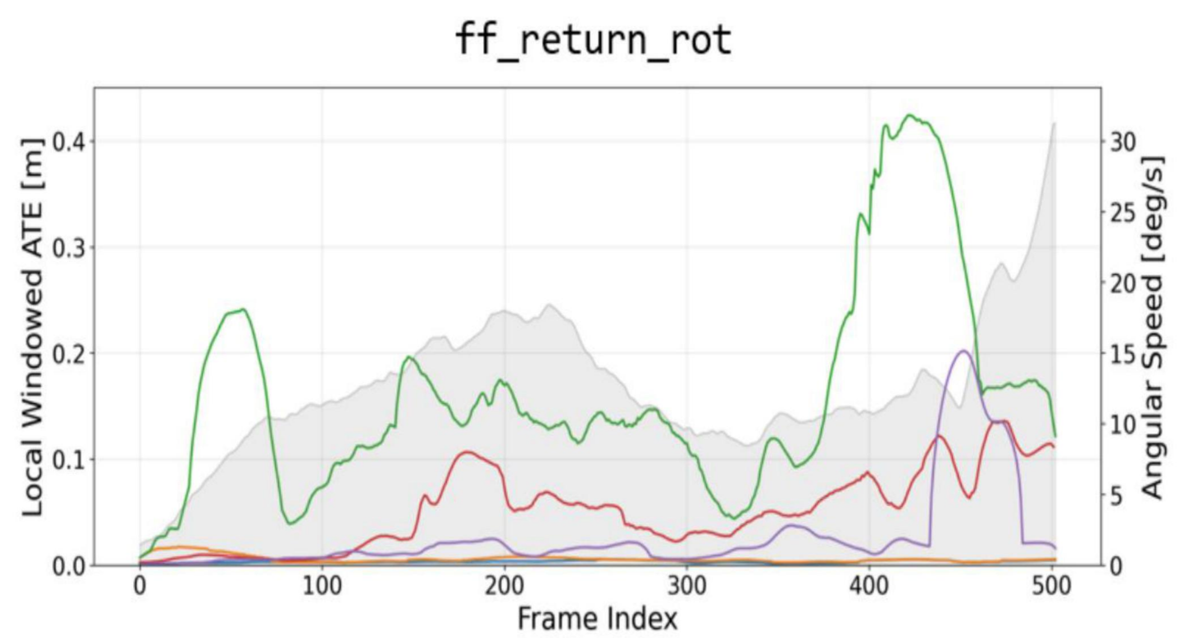
Method	iva_kibo_rot	ff_return_rot
HI-SLAM2 [16]	**0.013**	**0.006**
WildGS-SLAM [17]	0.087	0.015
MUSt3R [18]	0.169	1.001
MASt3R-SLAM [19]	0.196	0.682
MonoGS [15]	0.462	1.223

Key finding: Local robustness is motion-conditioned — errors spike during high-rotation intervals (gray).

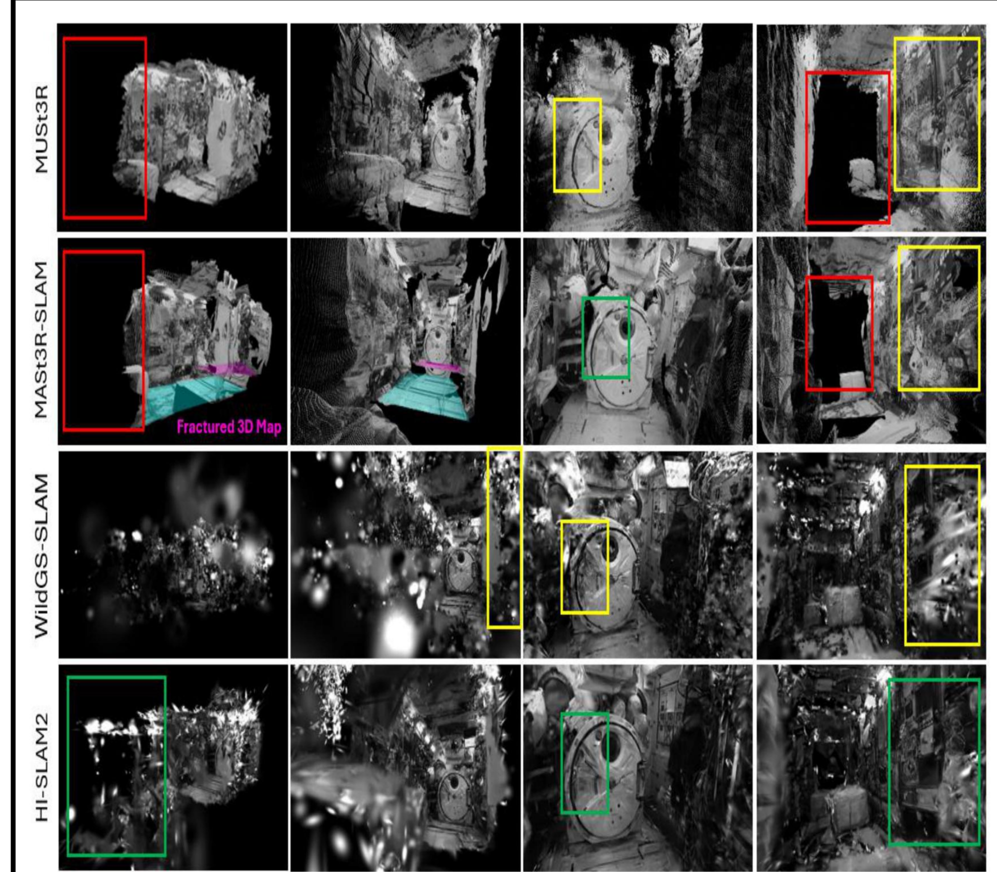
- **3DGS SLAM:** structural constraints and global optimization (e.g., HI-SLAM2) prevent tracking failure under extreme rotation.
- **Foundation models:** feed-forward priors alone produce local error spikes; temporal tracking mitigates them.

Local Windowed ATE vs. Angular Speed





Qualitative Results & Takeaways



1. **HI-SLAM2** (best)
2. WildGS-SLAM
3. MUSt3R
4. MASt3R-SLAM
5. **MonoGS** (collapses)

unreconstructed
floaters / blur
faithful recovery

Three failure modes under large rotation:

(i) MonoGS loses tracking under weak-overlap motion.
(ii) MUSt3R shows local spikes without temporal integration.
(iii) MASt3R-SLAM tracks well; loop closure fractures the map.

HI-SLAM2 — strongest local robustness and most coherent reconstruction.